

Highlights

PetroMind: A multimodal petrographic model for rock image classification and lithological description generation

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- A unified dataset is developed for instruction tuning in rock image classification and description
- A novel framework for rock image classification and description generation is constructed
- A novel LLM-based metric is developed to evaluate the semantic accuracy of lithological descriptions
- The collaborative fine-tuning of multimodal tasks demonstrates a synergistic effect

PetroMind: A multimodal petrographic model for rock image classification and lithological description generation

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ABSTRACT

Recent studies have revealed the remarkable capabilities of multimodal large models (MLLMs) in general vision-and-language tasks, generating increasing interest in their application to geoscientific domains. Rock image recognition and lithological description constitute fundamental skills for geologists and represent one of the earliest application scenarios in geosciences where artificial intelligence technologies have been actively explored. As rock image recognition and description are inherently multimodal tasks, involving both visual and textual data, models are required to understand the interactions between modalities and how these interactions affect information transfer. However, the simultaneous comprehension of rock images and their corresponding lithological descriptions remains a significant challenge for current machine learning frameworks. The impact of multimodal co-training on downstream geoscientific performance is still unconfirmed. This study presents PetroMind, a domain-adapted multimodal petrographic foundation model built upon Qwen2.5-VL, together with SA-Rock, a novel LLM-based metric designed to assess the semantic accuracy of rock image generation descriptions to evaluate the capabilities of MLLMs in petrographic multimodal tasks. In the rock image classification task, PetroMind achieves performance comparable to ViT-Base-Patch16-224, with accuracy and Macro-F1 scores exceeding or approaching 97%. This demonstrates PetroMind's strong capability in long-tailed image learning and highlights its effectiveness in substantially improving classification accuracy for few-shot rock image categories. In the lithological description generation task, PetroMind attains BLEU-4 and SA-Rock scores of 0.644 and 7.864, respectively, indicating good few-shot learning performance. The SA-Rock metric shows that the model produces highly accurate descriptions of rock structure, texture, and colour, while leaving considerable scope for improvement in the description of mineral composition. Ablation experiments further indicate that task-specific LoRA adapters are more effective for high-resource tasks such as image classification, whereas a single shared LoRA adapter demonstrates superior multi-task interaction in low-resource captioning scenarios. This study demonstrates the potential of MLLM-based architectures in jointly understanding rock images and their associated geological descriptions.

1. Introduction

Rocks are solid aggregates composed of minerals or mineral-like substances and constitute essential components of Earth and other terrestrial planets. Since the publication of the first geological map in 1815, lithological identification, the characterisation and mapping of surface and subsurface rock types, has remained a fundamental task in geology, underpinning geological surveys and mineral exploration (Fu et al., 2017). Over the past decade, emerging information technologies, particularly big data and artificial intelligence, have markedly transformed geoscience research (Zhou et al., 2021) and continue to reshape conventional workflows for lithological identification and description (Chen et al., 2022a; Chen et al., 2024). As one of the earliest domains in geosciences to adopt deep learning and related artificial intelligence (AI) techniques, intelligent lithological recognition and automated processing of lithological descriptions present a representative case for advancing AI applications in the discipline.

Rock images serve as a vital data source for lithological identification by geologists and autonomous exploration systems. Deep learning architectures, including AlexNet (Feng et al., 2019), VGG16 (Li et al., 2020b), Inception-V3 (Zhang et al., 2018), ResNet50 (Ren et al., 2021), and EfficientNet (Huang et al., 2023; Bai et al., 2025), have been applied to intelligent rock image classification at both thin-section and hand-specimen (including outcrop) scales.

However, current research is constrained by three major limitations:

(1) Dataset scale and completeness. Studies have shown that small training datasets often result in overfitting in geological image transfer learning (Dawson et al., 2023). Although existing datasets generally cover the three principal rock classes, igneous, sedimentary, and metamorphic, taxonomic inconsistency and incomplete coverage remain. For instance, several datasets include only granitic rocks under the igneous category, omitting syenites, diorites, gabbros, and other significant subtypes.

(2) Class imbalance. In addition to dataset size, class imbalance (Saporetti et al., 2018; Wang et al., 2024) persists as a critical challenge. Such an imbalance reduces model generalisation, making it difficult for models to learn new or underrepresented rock types with limited labelled samples.

(3) Underexplored multimodal petrographic tasks. Previous research on rock description has primarily addressed knowledge extraction and quality assessment (Chen et al., 2022b). Multimodal learning that simultaneously integrates rock images and their textual descriptions remains largely underdeveloped.

Since the release of ChatGPT in November 2022, numerous institutions worldwide have introduced their own general-purpose large language models (LLMs) (Kalyan, 2024; Annepaka and Pakray, 2024). These models have demonstrated remarkable generative capabilities in natural language processing (NLP). Building upon LLMs as backbones, MLLMs are increasingly attracting attention from both academia and industry. Within geosciences, researchers are actively evaluating the performance of such multimodal models. The concept of a multimodal foundation model for lunar and planetary science has recently been proposed (Lei et al., 2024). Existing studies indicate that MLLMs possess a substantial few-shot learning advantage on downstream tasks (Xiao et al., 2025). Rock image recognition and description are inherently multimodal tasks, requiring the integration of both visual and textual data. Effective models must capture interactions between modalities and demonstrate how these interactions influence information transfer. Nonetheless, the joint interpretation of rock images and their associated descriptive texts remains a substantial challenge for current machine learning frameworks. In addition, the influence of multimodal learning on model performance in geoscience-specific tasks remains largely unquantified and insufficiently explored.

This study develops PetroMind, a multimodal foundation model for petrography, built upon Qwen2.5-VL to assess the capabilities of MLLMs in petrographic tasks. First, paired rock image-text datasets designed for lithological recognition and description are constructed, enabling both single-task and multi-task training of the multimodal model. Specifically, to evaluate few-shot learning performance under class-imbalanced conditions, a comprehensive dataset of plutonic intrusive rocks is compiled, deliberately designed with imbalanced class distributions. Second, using these image-text pairs and the pre-trained Qwen2.5-VL backbone, the model is fine-tuned using low-rank adaptation (LoRA) to efficiently adapt the multimodal architecture to petrographic tasks, resulting in the final PetroMind model. Finally, comparative experiments are conducted under varying training strategies, including single-task versus multi-task learning and different LoRA configurations, to systematically evaluate performance in both rock image classification and caption generation tasks.

Accordingly, the contributions of this study can be outlined as follows:

(1) A rock image visual–language instruction-tuning dataset. One of the primary challenges in geoscientific multimodal modelling is the limited availability of high-quality, task-specific datasets. Building upon the curated, class-imbalanced dataset of plutonic intrusive rock images, this study develops dedicated instruction-tuning datasets for two core tasks: rock image classification and lithological description generation.

(2) PetroMind: the first multimodal foundation model for petrography. This study introduces PetroMind, the first petrography-specific multimodal model, to simultaneously address rock image recognition and lithological description produced by fine-tuning a vision-language foundation model (Qwen-VL) through LoRA.

(3) An automatic semantic evaluation metric for lithological descriptions. The study proposes SA-Rock, a novel evaluation framework that employs LLMs as evaluators. SA-Rock automatically generates both scores and interpretable rationales using chain-of-thought (CoT) prompting to assess the semantic accuracy of rock descriptions.

2. Related work

2.1. Rock image classification

Lithological identification typically requires systematic petrological knowledge and extensive practical experience with rock samples. Empirical findings reported by geological professionals indicate that even highly experienced field geologists achieve only 60–80% accuracy in identifying rock types under field conditions. Due to the rise of artificial intelligence, a fundamental challenge and opportunity in geosciences lies in translating expert geological knowledge into AI models capable of delivering intelligent lithological recognition services. Such tools will benefit not only professional geoscientists but also enthusiasts and non-specialists by enabling accurate rock identification (Li et al., 2020a). Given the significant differences between thin-section and hand-specimen (including outcrop) scale rock image analysis, this study concentrates primarily on the latter. Existing deep learning approaches for specimen-scale rock image classification can be broadly classified into three categories: 1) convolutional neural networks (CNNs), 2) vision transformers (ViTs), and 3) fusion and ensemble learning methods.

2.1.1 CNNs

In the early stages, relatively simple CNN architectures, such as AlexNet and VGG, were applied to the automated classification of granitic tiles (Ferreira and Giraldi, 2017). As CNNs advanced towards deeper architectures, models such as GoogLeNet (including Inception-V3) and ResNet were increasingly employed for rock image recognition. For instance, using a dataset of 15 common rock types, Inception-V3 achieved the highest reported accuracy of 63% (Bai et al., 2018). On a smaller dataset of three rock types, granite, phyllite, and breccia, a deep transfer learning model based on Inception-V3 achieved over 80% accuracy (Zhang et al., 2018). A custom model, RTCNN, demonstrated slightly superior performance compared to Inception-V3 on UAV-captured field images covering six rock types from igneous, sedimentary, and metamorphic categories: granite, conglomerate,

sandstone, shale, limestone, and mylonite (Ran et al., 2019). Likewise, on a core image dataset comprising 200 boxes across five rock classes, GoogLeNet delivered the strongest performance (Baraboshkin et al., 2020).

An interpretability study of deep transfer learning applied to plutonic intrusive rocks from the Dabie Mountains showed that deeper network architectures do not necessarily yield superior performance in rock image classification (Chen et al., 2023). Motivated by this finding, the study selects four representative CNN architectures, VGG16, Inception-V3, ResNet50, and DenseNet121, for comparative evaluation in subsequent classification experiments.

2.1.2 ViTs

Transformers, initially introduced by Vaswani et al. for machine translation, have since become state-of-the-art architectures across a broad spectrum of NLP tasks. Inspired by the success of scaling Transformers in NLP, researchers have adapted the architecture for direct application to image data. The vision transformer (ViT) (Dosovitskiy et al., 2021) represents a key model that applies Transformer mechanisms to image classification. ViT processes image patches as word tokens and is architecturally analogous to BERT, framing image classification as a “sentence-level” prediction task. It is recognised for its strong global modelling ability and rapid convergence. In geoscientific applications, researchers have also adopted pre-trained ResNet50 and Swin Transformer as backbone networks for thin-section image recognition (Wang et al., 2025).

When trained on mid-sized datasets such as ImageNet without robust regularisation, ViTs generally underperform relative to ResNets of comparable size by several percentage points. This trend was observed in comparative experiments across 11 mineral classes, where ViT consistently lagged behind ResNet101 both before and after data augmentation (Wang et al., 2024). However, this performance gap is reversed when ViTs are pre-trained on large-scale datasets (14M–300M images). When transferred to multiple mid- and small-scale image recognition benchmarks (ImageNet, CIFAR-100), ViTs not only match but often surpass CNNs, while also requiring fewer computational resources during fine-tuning. Multimodal models based on ViT are typically pre-trained on extensive image–text paired datasets and fine-tuned for downstream tasks. In addition, multimodal long-tail learning is emerging as a promising research direction (Han et al., 2025). In rock image classification, where many lithological classes suffer from insufficient sample sizes, integrating MLLMs can provide supplementary supervision to underrepresented classes, enhancing overall model robustness and generalisation.

2.1.3 Fusion and Ensemble Learning Methods

As performance improvements from single-architecture models begin to plateau, ensemble learning strategies have been increasingly adopted to advance rock image recognition. Early studies employed Siamese network architectures, combining distinct backbone models to extract and integrate multi-scale features from rock images (Feng et al., 2019). Fine-tuning experiments on a Mars surface rock dataset (6,691 images across 25 classes) demonstrated that although individual ImageNet-pretrained models produced suboptimal results, ensemble methods effectively combined multiple predictions to achieve superior accuracy (Nandi et al., 2022). Wang et al. (2024) employed DCGANs to synthesise images for underrepresented mineral classes, enhancing model performance to address class imbalance in mineral image datasets. They integrated multiple CNN and Transformer models to improve overall classification accuracy. More recently, Zhu et al. (2025) introduced RockNet, a deep progressive learning model designed to identify complex lithological patterns.

Although ensemble and generative approaches enhance performance on imbalanced datasets, they remain limited in their ability to exploit semantic textual knowledge, a gap addressed by the multimodal instruction-tuning framework PetroMind.

2.2. Lithological description generation

Traditionally, lithological description generation has been conducted manually by geologists, who visually examine field outcrops, hand specimens, or thin sections under a microscope to document characteristics such as colour, texture, structure, and mineral composition. However, this process is inherently subjective, heavily dependent on individual expertise, and time-consuming. Rock descriptions are typically concise and structured, which makes them well-suited to NLP tasks. Previous research has addressed knowledge extraction (Chen et al., 2022a) and quality assessment (Chen et al., 2022b) of such texts. More recently, automated lithological description generation has begun to attract interest in the geoscience community. For instance, Paucar et al. (2025) trained a hybrid deep learning model using an image-caption dataset. Their method employed EfficientNetB7 to extract visual features and a Transformer decoder to generate textual descriptions, with outputs converted into speech via text-to-speech synthesis. Using accuracy and BLEU as evaluation metrics, the model achieved scores of 0.892 and 0.71, respectively, which indicates promising performance. Given the generative capabilities of LLMs in NLP, MLLMs, capable of jointly processing images and text, have naturally emerged as a compelling direction for automating lithological description generation in geosciences.

2.3. MLLMs

In recent years, breakthroughs in LLMs (Annepaka and Pakray, 2024) and large vision models (LVMs) (Wang et al., 2023) have accelerated the development of MLLMs, architectures capable of jointly processing visual inputs, understanding language, and performing reasoning (Yin et al., 2024). These models are now at the forefront of AI research. Within geoscientific applications, researchers have utilised MLLMs through Microsoft Azure OpenAI services to design GeoMap-Agent, a domain-specific intelligent agent intended to enhance the accuracy and efficiency of geological map interpretation (Huang et al., 2025). Table 1 summarises two representative MLLMs: LLaVA (Liu et al., 2023a; Liu et al., 2024) and Qwen-VL (Bai et al., 2025). LLaVA represents the first end-to-end framework for vision-language instruction tuning, featuring a streamlined architecture that connects a CLIP-ViT visual encoder with the Vicuna LLM. In contrast, Qwen-VL is pre-trained on hundreds of millions of image-text pairs and extends the Qwen-7B LLM with visual capabilities through integration with a ViT encoder and a position-aware visual grounding module.

Table 1. Comparison of typical MLLMs.

Item	LLaVA-1.5	Qwen2.5-VL
Vision encoder	CLIP-ViT	OpenCLIP-ViT
Image resolution	224×224-336×336	Native dynamic resolution
Modality alignment method	MLP	MLP
LLM backbone	Vicuna v1.5	Qwen2.5
Pre-training data scale	558K public image-text pairs + GPT-4 synthetic data, with less than 5% in Chinese	1.2M public image-text pairs + Alibaba's proprietary data, with 38% in Chinese
Training stage	Two-stage: language-image pre-training, instruction fine-tuning	Three-stage: language-image pre-training, multi-task joint training, instruction fine-tuning

2.3.1 Vision encoder

The vision encoder V in MLLMs plays a critical role in transforming raw visual inputs I_x into compact visual feature representations Z_x (Equation 1). Instead of training encoders from scratch, it is standard practice to adopt pre-trained vision encoders, particularly those trained on large-scale image-text datasets. A prominent example is contrastive language-image pre-training (CLIP), introduced by OpenAI in 2021 (Radford et al., 2021), which has become the de facto standard for vision-language alignment in MLLMs.

$$Z_x = V(I_x) \quad (1)$$

CLIP leverages text, which is richer and more expressive than traditional visual labels, to train its encoder, enabling zero-shot learning across a wide range of vision–language tasks, including image classification, captioning, and visual question answering. Its multimodal design ensures robust semantic alignment between images and text, providing a foundation for comprehensive cross-modal understanding in diverse applications. CLIP supports both ResNet and ViT as backbone architectures. Empirical evaluations on ImageNet-Real (real-world images) and ImageNet-V2 (challenging test sets) demonstrate that ViT consistently outperforms ResNet in representation quality and generalisation. Based on this advantage, both LLaVA and the Qwen-VL series adopt ViT as their visual backbone.

2.3.2 LLM backbone

MLLMs typically employ an LLM as the decoder to perform text generation and reasoning tasks. In Equation (2), L is the LLM decoder, T_x is the embedded token sequence of the input text, and H_x is the visual feature representation aligned to the LLM’s semantic space. Compared to closed-source models such as GPT-4, open-source LLMs, including LLaMA and Qwen, provide a foundational platform for MLLM research, enabling full model access, reproducibility, and task-specific customisation. Given the documented performance disparities of LLMs across languages, particularly in handling domain-specific Chinese geological terminology, this study adopts the Qwen series as the decoder backbone owing to its native optimisation for Chinese language understanding and generation.

$$R = L(H_x, T_x) \quad (2)$$

2.3.3 Modality alignment

The visual feature vectors Z_x produced by the vision encoder are not inherently interpretable by the language decoder of an LLM. A modality alignment mechanism, denoted as f , is therefore required to transform Z_x into a semantically aligned representation H_x that the LLM can effectively process. Qwen-VL enhances spatial awareness through a high-capacity vision encoder and dynamic attention mechanisms, making it particularly suitable for fine-grained visual tasks. In contrast, LLaVA employs a lightweight design, applying a single-layer MLP as its modality alignment module, prioritising computational efficiency and rapid inference.

$$H_x = f(Z_x) \quad (3)$$

Although existing MLLMs can follow complex instructions with considerable accuracy, their capacity to solve domain-specific problems, particularly in medicine, remains limited, and they continue to be prone to generating hallucinations or factually incorrect outputs (Xiao et al., 2025). In geosciences, where precise lithological reasoning and domain knowledge are critical, task-specific instruction tuning is indispensable to adapt general-purpose MLLMs for reliable and accurate performance.

3. Datasets and method

3.1. Datasets

The instruction tuning dataset is divided into two components based on the objectives of the multimodal training tasks: 1) instructions for the rock image classification task, and 2) instructions for the lithological description generation task. The classification instructions are derived by converting an existing CNN-based rock image dataset, preserving the original labels and enriching them with standardised classification prompts. Due to cost constraints, the captioning instructions are constructed by randomly sampling images per rock class and manually composing expert-validated lithological descriptions for each image. All instruction–response pairs are organised in OpenAI’s ShareGPT format and stored as JSON files to ensure compatibility with widely used instruction-tuning frameworks.

3.1.1 Rock image dataset

Focusing on plutonic intrusive rocks, ROCK-13 is constructed as a class-imbalanced rock image dataset intentionally designed to reflect a long-tailed distribution. The dataset includes the following 13 lithological classes: olivine, pyroxene-hornblende, gabbro, diabase, anorthosite, diorite, monzonite, syenite, granodiorite, monzonitic granite, syenogranite, alkali feldspar granite, and tonalite. Previously, following Soviet-era nomenclature, tonalite was referred to as "plagiogranite" (Chang et al., 2009). However, under the IUGS classification scheme for plutonic rocks, the term "plagiogranite" has been deprecated. In addition, pyroxenite and hornblendite are also merged into a single class (“pyroxene-hornblende rock”) due to the difficulty of distinguishing them at the hand-specimen scale.

Rock images were collected from field outcrops and museum collections, including specimens from the Cores and Samples Center of Natural Resources, China Geological Survey, using consumer-grade smartphones commonly employed by geologists (iPhone 12, Huawei Mate 60). After screening, preprocessing, and expert annotation, images were sampled per class based on five tiers of data volume: 15, 50, 200, 500, and 1,000 images, respectively, corresponding to approximate ranges of 15–50, 50–200, 200–500, 500–1,000, and >1,000. This process yielded a

final dataset of 5,845 manually annotated images across 13 plutonic rock types. Basic statistics of data collection and annotation are listed in Table 2, while example images for each rock class are shown in Figure 1.

Table 2. Basic statistics of data collection and annotation in the ROCK-13 dataset.

No.	Rock type	Number collected	Number of classification labelled	Number of caption labelled
1	Olivine	58	50	13
2	Pyroxene-hornblendite	621	500	36
3	Gabbro	700	500	23
4	Diabase	242	200	10
5	Anorthosite	19	15	10
6	Diorite	971	500	20
7	Syenite	1053	1000	30
8	Monzonite	506	500	20
9	Granodiorite	1322	1000	30
10	Monzonitic granite	1095	1000	30
11	Syenogranite	864	500	20
12	Alkali feldspar granite	15	15	10
13	Tonalite	51	50	20

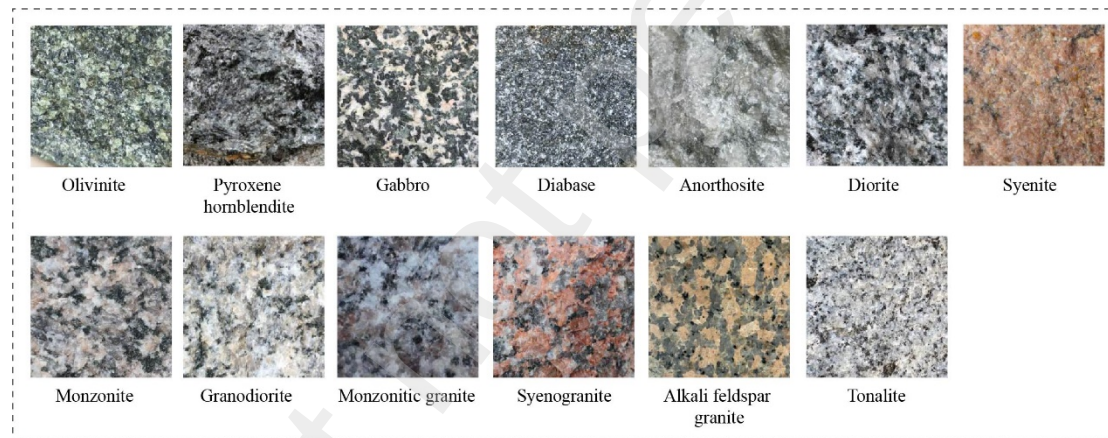


Fig. 1. Examples of rock image dataset.

3.1.2 Instruction Tuning Datasets

Two distinct sets of supervised instruction tuning data were constructed to support both rock image classification and lithological description generation. For the classification task, geometric data augmentation techniques, including rotation and horizontal flipping, were applied to the original 5,845 annotated images, generating a high-quality expanded set of 58,450 augmented samples. Each image was then paired with a

standardised instruction-response template, resulting in 58,450 classification-oriented instruction pairs. For the lithological description generation task, a small number of representative images per rock class were randomly selected for manual annotation by domain experts, producing 272 high-fidelity image-text pairs (Table 2). For both tasks, instruction tuning data were divided into training and validation sets at a 4:1 ratio.

3.2. The PetroMind Framework

Figure 2 indicates that the PetroMind multimodal framework consists of three principal components: the ViT-based vision encoder, the vision-text alignment module (based on CLIP), and the LLM-based decoder for text generation and reasoning.

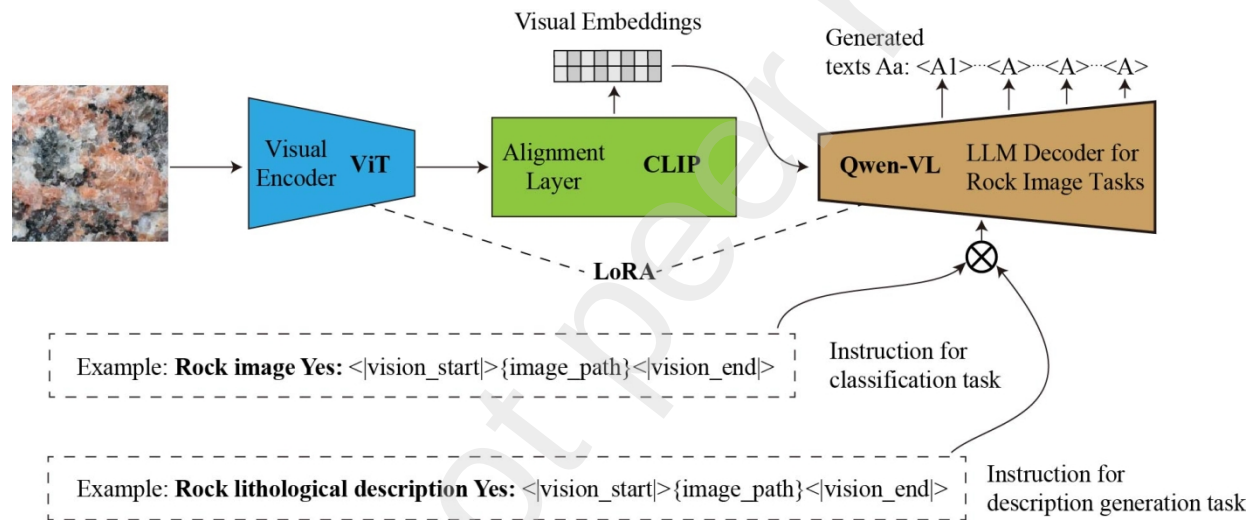


Fig. 2. Multimodal learning framework of the proposed PetroMind.

The Qwen2.5-VL model (Bai et al., 2025) is adopted as the underlying multimodal architecture, initialised with pre-trained weights from Qwen2.5. The vision encoder applies OpenCLIP-ViT-bigG-14, which partitions input rock images into fixed-size patches (14×14 pixels) and employs windowed attention mechanisms. This design natively supports dynamic input resolutions, a feature that is particularly important for managing geological images with diverse spatial scales. The vision-text alignment module incorporates a lightweight MLP-based strategy: it groups spatially neighbouring sets of four patch features and projects them through a two-layer MLP into a dimension aligned with the LLM's text embedding space, facilitating efficient cross-modal fusion.

Although large-scale pre-training followed by domain adaptation has become a prevailing paradigm for deploying general-purpose LLMs across disciplines, full fine-tuning, which updates all model parameters, is increasingly impractical for models such as LLaVA or Qwen-VL when computational budgets are limited. Therefore, LoRA (Hu et al., 2022) is employed as a parameter-efficient method that freezes the pre-trained weights while introducing trainable low-rank decomposition matrices into each Transformer layer. Within the PetroMind framework, LoRA is integrated into both the vision encoder and the LLM decoder, as follows:

- In the ViT encoder, LoRA branches are embedded into each Transformer layer to augment the model's ability to extract discriminative visual features from rock images.
- In the LLM decoder, LoRA is applied to the query (Q), key (K), and value (V) projection matrices of the multi-head self-attention layers, enabling the model to refine its linguistic style in line with geological terminology and lithological description conventions.

This approach reduces the number of trainable parameters by more than 95% compared to full fine-tuning, lowering GPU memory requirements and training time, while maintaining task performance.

4. Experiment

4.1. Evaluation Methods

4.1.1 Evaluation metric for the rock image classification task

Commonly employed evaluation metrics in multi-class classification tasks include accuracy and the *F1-score*. Accuracy provides an intuitive and computationally efficient indicator of overall performance, particularly when class distributions are balanced and misclassification costs are uniform. However, in the presence of class imbalance, which is a frequent characteristic of geoscientific datasets, the *F1-score* is typically favoured to counter bias towards majority classes.

The F1-score is usually reported in two primary forms: macro-averaged F1-score (*Macro-F1*) and micro-averaged F1-score (*Micro-F1*). *Macro-F1*, which computes the arithmetic mean of per-class F1-scores, treats all classes equally, irrespective of their sample sizes. This property makes it particularly effective for assessing performance on minority classes, providing a more equitable evaluation in imbalanced datasets (Sokolova and Lapalme, 2009). This study selects *Macro-F1* as the primary evaluation metric to highlight the model's capacity for few-shot or low-resource class learning. The computation of *Macro-F1* is formally expressed in Equation (4).

$$Macro - F1 = \frac{1}{N} \sum_{i=1}^N F1_i \quad (4)$$

4.1.2 Evaluation metric for the lithological description generation task

The evaluation of text generated by LLMs continues to present considerable challenges. Conventional metrics such as BLEU and ROUGE have demonstrated relatively weak correlations with human judgements of semantic fidelity and contextual appropriateness (Liu et al., 2023b). Therefore, recent work has explored the use of LLMs themselves as evaluators, employing CoT prompting to generate both scores and justifications. In this study, the evaluation focuses on the semantic accuracy of lithological descriptions produced from rock images. An automated evaluation pipeline is implemented using the StringEvaluator module from LangChain (v0.3.26), configured to compare generated texts against predefined geological criteria.

A domain-specific metric termed semantic accuracy for rock image generation descriptions (SA-Rock) is introduced to enhance consistency and interpretability in LLM-based evaluations. The SA-Rock score is expressed on a 10-point scale, with higher values reflecting closer semantic alignment with expert-curated reference descriptions. Table 3 indicates that the scoring framework emphasises two central dimensions: (1) semantic proximity to the reference answer, and (2) completeness of geological information conveyed (mineralogy, texture, colour, structure). The SA-Rock evaluation is decomposed into thematic subcomponents to improve interpretability and facilitate human validation:

- SA-Rock_{color} is the accuracy in describing rock colour.
- SA-Rock_{mineral} is the fidelity in identifying and describing mineral constituents.
- SA-Rock_{texture} is the correctness in characterising grain size and fabric.
- SA-Rock_{structure} is the precision in depicting macroscopic structures (bedding, foliation).

Each subcomponent is evaluated using a dedicated CoT prompt that is explicitly designed to guide the LLM through a stepwise, attribute-specific reasoning process grounded in geological principles.

The automated assessment proceeds in two stages:

- Thematic Extraction: the LLM first parses the generated description to isolate segments corresponding to each geological theme (colour, mineral, texture, structure);
- Thematic Scoring: for each extracted segment, a second LLM instance evaluates its semantic accuracy against the reference by applying the corresponding SA-Rock submetric.

Equation (5) shows that the final SA-Rock score is calculated as the arithmetic mean of all thematic sub-scores, providing a holistic measure of semantic fidelity for the complete lithological description. For comparative benchmarking, all generated texts are also evaluated using BLEU-4, enabling cross-validation between traditional n-gram metrics and the proposed LLM-based approach. As an additional measure, human evaluation is conducted by domain experts (geologists with field and petrographic experience), who independently rate description accuracy using the same 10-point SA-Rock rubric. Human judgements are then employed to validate and calibrate the automated scores, ensuring geological relevance and robustness.

$$SA-Rock = \frac{1}{N} \sum_{i=1}^N SA-Rock_i \quad (5)$$

Table 3 Examples of the prompt words

No.	Subject	Prompt content snippet
1	Color	分数 7: 答案中出现与颜色相关的描述, 所描述的颜色与参考文献中的颜色色系相近。 (Score 7: The answer includes descriptions related to colour, and the described colour is similar to the colour range provided in the references.)
		分数 10: 答案中出现与颜色相关的描述, 颜色描述与参考文献完全一致。 (Score 10: The answer includes descriptions related to colour, and the colour description is entirely consistent with the references.)
2	Mineral	分数 5: 答案中出现与矿物成分相关的描述, 在矿物成分描述细节上与参考文献有相关性, 但存在不准确之处。 (Score 5: The answer includes descriptions related to mineral composition, and while there is some relevance to the details in the reference material, inaccuracies are present.)
		分数 7: 答案中出现与矿物成分相关的描述, 在矿物成分描述细节上大部分与参考文献一致, 但有一些小错误或遗漏。 (Score 7: The answer includes descriptions related to mineral composition, with most details consistent with the reference material, but contains minor errors or omissions.)

4.2. Experimental Setup

Most experiments were executed in Python using the PyTorch framework on a workstation equipped with two NVIDIA RTX 4090 GPUs. For experiments requiring greater computational resources, a Linux-based container environment was deployed on a high-performance computing (HPC) platform powered by two NVIDIA A40 GPUs. Detailed specifications of both experimental environments are provided in Table 4. Pretrained parameters for ViT and multimodal models were obtained from ModelScope on the workstation and transferred to the Linux container. All experiments involving CNNs were performed locally on the workstation, with pretrained weights accessed through TorchVision. For parameter-efficient fine-tuning, the PEFT library (v0.15) was employed to implement LoRA. The LoRA configuration is detailed in Table 5: the learning rate was set to $1e-4$, the task type was

configured as causal language modelling (CAUSAL_LM), and target modules for adaptation were specified as ["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"].

Table 4 Experimental environment

Experimental environment	Windows workstation	Linux container
GPU	NVIDIA RTX 4090 * 2	NVIDIA A40 * 2
Programming language	Python 3.11.11	Python 3.11.5
Deep learning framework	PyTorch 2.6.0	PyTorch 2.1.1
CNN model library	Torchvision 0.21.0	Torchvision 0.16.1
LLM open-source library	ModelScope 1.24.1	ModelScope 1.26.0
Fine-tuning library	PEFT 0.15.1	PEFT 0.15.2
CUDA version	12.6	12.0

Table 5 LoRA fine-tuning parameters

Task type	Alaph	Dropout	lr	Bias	Target modules
CAUSAL_LM	16	0.05	1e-4	none	["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"]

References: lr: learning rate.

4.3. Experimental results

The pre-trained MLLM utilised in the experiments is Qwen2.5-VL-7B-Instruct. For LoRA fine-tuning, task-specific rank values were selected to optimise adaptation efficiency: rank = 64 for the rock image classification task and rank = 128 for the lithological description generation task. A detailed discussion of rank selection strategies across different tasks is provided in Section 5.

The fine-tuning training curves demonstrate a stable and effective optimisation process: the loss decreased rapidly during early training steps. At the same time, the gradient norm (grad_norm) exhibited a steady and consistent decline, consistent with typical convergence behaviour observed in standard deep learning models. Final fine-tuning results for the PetroMind model are listed in Table 6. On the rock image classification testing dataset, PetroMind achieved a Macro-F1 score of 97.35%, indicating high discriminative accuracy across all rock classes. On the caption generation testing dataset, it attained an average SA-Rock score of 7.864, indicating strong semantic alignment between generated descriptions and ground-truth annotations.

Table 6 Experimental results

Pre-trained MLLM	Image size	Batch size	Classification		Captioning	
			rank	Macro-F1	rank	SA-Rock
Qwen2.5-VL-7B-Instruct	224 × 224	8	64	97.35	128	7.864

5. Discussion

5.1. Impact of Training and Sampling Strategies on Classification Task Performance

Performance in the rock image classification task is typically influenced by factors such as model architecture, training strategy, and sampling methodology. Table 7 exhibits that CNNs are generally adapted to domain-specific tasks either through transfer learning from ImageNet-pretrained weights or by training from scratch. Models that achieve progressively higher performance on ImageNet also demonstrate consistent gains on the ROCK-13 dataset, indicating strong cross-domain transferability. DenseNet121, benefiting from both architectural optimisation and ImageNet-based pre-training, achieved the highest classification accuracy of 97.15%. However, the impact of class imbalance remains evident: across all evaluated models, macro-averaged F1-scores are consistently lower than overall accuracy scores, highlighting the challenge of maintaining balanced performance across underrepresented rock classes.

Table 7 Transfer learning parameters and results of the typical CNNs

Model	Type	FLOPs (G)	Params (M)	Epoch	Time (H:M:S)	Accuracy	Macro-F1
VGG16	scratch	1979.68	134.31	200	11:21:37	74.52	69.97
VGG16	pre-training	1980.20	138.37	200	11:30:09	94.61	66.18
Inception-V3	scratch	365.64	23.83	200	14:35:08	90.20	86.13
Inception-V3	pre-training	365.64	23.85	189	14:01:14	94.10	66.07
ResNet50	scratch	528.86	23.53	200	07:49:19	90.74	<u>87.17</u>
ResNet50	pre-training	529.12	25.57	200	07:49:43	<u>96.35</u>	82.03
DesNet121	scratch	370.69	6.97	132	07:14:18	91.16	90.79
DesNet121	pre-training	370.82	7.99	147	07:44:41	97.15	82.51

References: The **bold** denotes the highest result, and the underline denotes the second-highest result.

Class re-sampling is a widely adopted data-handling strategy to reduce the negative impact of class imbalance. In particular, the weighted random sampler assigns higher sampling probabilities to underrepresented classes, increasing their presence during training and raising more balanced optimisation. Table 8 exhibits that after applying the weighted random sampler, the performance gap between *Macro-F1* and overall accuracy is substantially reduced across all models. ViT-based architectures, including the proposed PetroMind model (which employs ViT as the

visual encoder), exhibit significantly lower sensitivity to sampling strategies compared to traditional CNNs. This architectural robustness indicates that ViT-based models are better able to learn discriminative features without heavily relying on class-balanced sampling. In addition, the PetroMind model achieves performance comparable with that of ResNet50, DenseNet121, and ViT-Base-Patch16-224, with all models attaining *Macro-F1* scores exceeding or closely approaching 97%.

Table 8 Parameters and results for transfer learning and LoRA fine-tuning of the PetroMind

Model	Type	Sampling	Params (M)	Epoch	Time (H:M:S)	Accuracy	Macro-F1
ResNet50	scratch	Yes	23.55	102	04:00:07	89.69	88.60
ResNet50	pre-training	Yes	25.57	74	02:55:59	96.87	96.94
DesNet121	scratch	Yes	6.98	117	06:36:27	91.70	90.47
DesNet121	pre-training	Yes	7.99	188	10:07:29	97.75	97.94
ViT-Base-Patch16-224	pre-training	No	85.82	200	12:54:46	98.25	<u>98.13</u>
ViT-Base-Patch16-224	pre-training	Yes	85.82	200	13:15:05	<u>98.23</u>	98.60
PetroMind	LoRA	No	90.77	5	19:38:49	97.35	97.35

References: The **bold** denotes the highest result, and the underline denotes the second-highest result.

Figure 3 illustrates the confusion matrices of different models evaluated on the ROCK-13 testing dataset. The VGG16 model trained from scratch demonstrates reasonably strong classification performance across most rock classes; however, its accuracy declines for minority classes, highlighting its susceptibility to class imbalance. In contrast, the ResNet50 model, also trained from scratch but with a weighted random sampler, delivers substantially improved performance, particularly for underrepresented categories, confirming the effectiveness of sample re-weighting in mitigating imbalance. The confusion matrices of the proposed PetroMind model and the ViT-Base-Patch16-224 model are nearly identical, with both achieving consistently high per-class accuracy across all rock types and showing no pronounced bias towards particular categories. This outcome indicates that transformer-based architectures, especially when fine-tuned with domain-relevant multimodal data, are inherently more robust to class distribution skew, making class re-sampling strategies less essential for sustaining high classification fidelity.

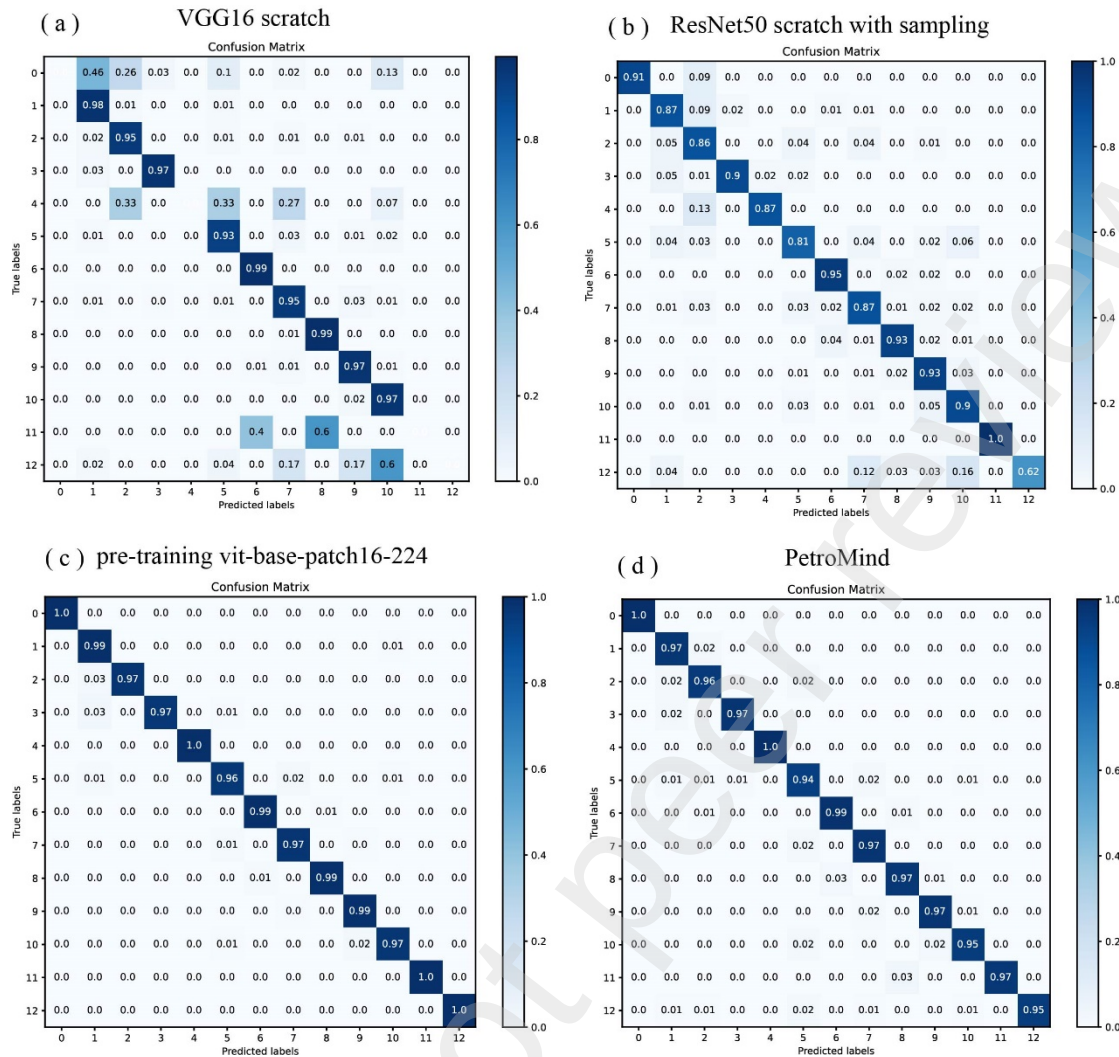


Fig. 3. Confusion matrices of model predictions on the testing dataset of ROCK-13. (a) The confusion matrix of the VGG16 model trained from scratch. (b) The confusion matrix of the ResNet50 model training from scratch and using the weighted random sampler. (c) The confusion matrix of the pre-training ViT-Base-Patch16-224 model. (d) The confusion matrix of the proposed PetroMind.

5.2. Comparison of Models for Geological Text Generation

Lithological description generation, a canonical multimodal task bridging computer vision and NLP, has traditionally been approached using architectures such as EfficientNetB7+Transformer for automated captioning of thin-section rock images. This study conducts a direct comparative evaluation between the proposed PetroMind model and a fine-tuned EfficientNetB7+Transformer baseline, with results listed in Table 9. The comparative analysis demonstrates that, despite the extreme data scarcity inherent in this setup (only 272 image-text pairs for

instruction tuning, constituting a genuine few-shot learning scenario), PetroMind consistently surpasses the baseline in both BLEU-4 and SA-Rock scores, exhibiting superior semantic grounding and geological coherence under low-resource conditions.

Table 9 Performance comparison of different methods on the lithological description generation task

Model	BLEU-4	Color	Texture	Stucture	Mineral	SA-Rock
EfficientNetB7+Transformer	0.103	7.444	3.074	2.796	1.852	3.792
PetroMind	0.644	8.655	8.127	8.964	5.709	7.864

References: The **bold** denotes the highest result.

A human evaluation study was conducted in parallel with automated scoring across several descriptive dimensions, and the root mean square error (RMSE) was computed to quantify alignment to assess the effectiveness of the proposed evaluation metric for geological caption generation. Table 10 indicates that the results for the PetroMind model across different target attributes reveal a strong agreement between automatic and human ratings, confirming the reliability of the SA-Rock metric.

However, manual inspection exposed several systematic shortcomings in the generated descriptions: (1) in terms of colour, lower-scoring outputs often correctly identified the fresh rock colour but neglected weathered surface hues; (2) with respect to texture and structure, the model frequently failed to recognise specific petrographic terms such as “gabbroic” or “granitic texture”, defaulting instead to generic phrases such as “granular structure”; and (3) regarding mineral composition, scores typically ranged between 5 and 7 (on a 0–10 scale), with five assigned when mineral types were misidentified and 7 when errors occurred in specifying modal abundance or the degree of euhedral development. These findings indicate that, although PetroMind-generated descriptions successfully capture the essential geological components (colour, texture, structure, and mineralogy), fine-grained accuracy within each category remains constrained, necessitating future improvements in both the quantity and quality of training data to strengthen applicability in geological fieldwork and laboratory contexts.

Table 10 Evaluation results across different target objects

Object	SA (Auto)	Manual	RMSE
Color	8.655	8.745	0.842
Texture	8.127	8.327	0.674

Structure	8.964	8.909	0.405
Mineral	5.709	6.291	1.321

5.3. Impact of Model Scale on Performance in Classification and Text Generation Tasks

Understanding the scaling behaviour and transfer properties of language models has become increasingly important in the evolution of large foundation models. In 2020, OpenAI introduced the Scaling Laws for Neural Language Models (Kaplan et al., 2020), which empirically demonstrated that model loss decreases based on a power-law trend as model size, dataset volume, and computational budget increase, indicating that performance generally improves predictably with scale. Subsequent work by Hoffmann et al. (2022) established that, under a fixed compute budget, optimal performance is achieved when model size and training data are scaled proportionally. However, it remains unresolved whether these scaling laws generalise to domain-specific downstream tasks in the geosciences. Table 11 indicates that the performance of multimodal models with varying parameter scales (3B, 7B, and 32B) was evaluated on both rock image classification and lithological description generation tasks. All comparative experiments employed task-specific LoRA adapters with a fixed rank of 32 to minimise confounding factors. The results indicate that scaling from 3B to 7B consistently enhances performance across both tasks. However, further scaling to 32B does not sustain these gains; *SA-Rock* performance regresses to the level observed with the 3B model. This finding indicates that, for geoscientific applications, a moderately scaled multimodal model (7B parameters) can provide the optimal balance between efficiency and effectiveness for downstream fine-tuning. In addition, enhancing task performance necessitates expanding training data volume in approximately proportion to model scale.

Table 11 Performance comparison of models with different parameter scales with task-specific LoRA adapters

Model	Classification (r=32)		Caption (r=32)				
	Accuracy	Macro-F1	Color	Texture	Structure	Mineral	SA-Rock
Qwen2.5-VL-3B-Instruct	93.10	92.60	8.182	7.964	9.327	4.964	7.609
Qwen2.5-VL-7B-Instruct	97.28	97.29	8.582	8.109	9.091	5.545	7.832
Qwen2.5-VL-32B-Instruct	95.60	95.03	8.600	7.164	9.455	5.509	7.682

References: The **bold** denotes the highest result; r: the rank of LoRA.

5.4. Ablation Study

In this section, the impact of different LoRA adapter configurations and hyperparameters within the PetroMind model is examined.

LoRA Adapters. Comparative experiments were performed to investigate the effect of using a single shared LoRA adapter versus task-specific LoRA adapters on model performance. The results are listed in Tables 12 and 13. The findings reveal that models with task-specific LoRA adapters consistently outperform those employing a single shared adapter in the rock image classification task: the lowest macro-F1 score achieved with the task-specific configuration (97.10) surpasses the highest score achieved with the shared-adapter setup (96.67). In contrast, in the lithological description generation task, the shared LoRA adapter delivers the best performance, reaching an SA-Rock score of 7.864.

Prior studies have indicated that task identifiers enable LLMs to distinguish between tasks, allowing a single LoRA adapter to facilitate cross-task parameter sharing and improve multi-task learning efficiency. However, other works highlight the risk of task interference in multi-task settings, where joint training can degrade rather than enhance performance, requiring mechanisms such as task routers to isolate task-specific computations. The instruction-tuning dataset for classification contains 58,450 samples, vastly exceeding the 272 image-text pairs available for captioning. Within this data regime, the shared LoRA adapter improves performance on the low-resource captioning task. However, it fails to enhance performance on the high-resource classification task.

Low-Rank Adaptation. In LoRA fine-tuning, the rank parameter governs both the number of trainable parameters introduced and the representational capacity of the adapter. A lower rank implies fewer trainable parameters and reduced flexibility of the model. Systematic ablation studies on rank selection were conducted, evaluating both task-specific adapters (Table 12) and the single shared LoRA adapter (Table 13). When task-specific LoRA adapters are employed, evaluation metrics exhibit an inverted U-shaped pattern: performance initially improves with increasing rank, peaks at an intermediate value, and then declines, consistent with theoretical expectations (Hu et al., 2022). Ranks that are too low restrict the model's ability to capture complex visual or linguistic patterns essential for classification or captioning, whereas excessively high ranks introduce overfitting risks. In contrast, with the single shared LoRA adapter, performance on both captioning and classification tasks shows a near-monotonic improvement as rank increases. This indicates that multi-task learning under parameter sharing demands greater representational capacity, with more trainable parameters required to manage cross-task complexity and mitigate interference. In addition, as detailed in Table 10, the RMSE between human and automatic

SA-Rock scores generally hovers around 1.0. Although the absolute gains observed across different rank settings fall within this margin of error, the consistent directional trends across both automatic and manual evaluations provide empirical support for the discriminative strength and reliability of the SA-Rock metric.

Table 12 Ablation on the rank of LoRA in PetroMind with task-specific LoRA adapters

rank	Color	Texture	Structure	Mineral	SA-Rock	Accuracy	Macro-F1
16	8.691	8.000	9.036	<u>5.491</u>	<u>7.805</u>	<u>97.34</u>	97.10
32	<u>8.582</u>	<u>8.109</u>	9.091	5.545	7.832	97.28	97.29
64	8.418	8.145	<u>9.182</u>	5.473	<u>7.805</u>	97.35	<u>97.35</u>
96	8.564	7.891	9.273	5.055	7.696	97.23	97.50
128	8.345	8.000	9.273	5.273	7.723	97.26	97.29

References: The **bold** denotes the highest result, and the underline denotes the second-highest result.

Table 13 Ablation on the rank of LoRA in PetroMind with a single shared adapter

rank	Color	Texture	Structure	Mineral	SA-Rock	Accuracy	Macro-F1
16	7.255	7.236	8.127	5.036	6.914	95.74	94.34
32	<u>8.491</u>	7.873	8.909	5.545	7.705	95.91	96.02
64	8.418	7.800	8.818	5.600	7.659	<u>96.12</u>	<u>96.05</u>
96	8.418	<u>7.945</u>	9.273	<u>5.655</u>	<u>7.823</u>	95.79	95.89
128	8.655	8.127	<u>8.964</u>	5.709	7.864	96.56	96.67

References: The **bold** denotes the highest result, and the underline denotes the second-highest result.

6. Conclusion

This study introduces PetroMind, a large multimodal foundation model specifically designed for petrographic analysis, instruction-tuned on domain-specific datasets covering both rock image classification and lithological description generation tasks in response to the multimodal nature of rock image recognition and geological captioning. It proposes SA-Rock in parallel as a novel LLM-based evaluation metric developed to quantify the geological fidelity and semantic correctness of generated descriptions. Through systematic comparative and ablation experiments examining the influence of training and sampling strategies, model scale, and architectural design, the results show that: (1) On the rock image classification task, PetroMind achieves performance comparable to ViT-Base-Patch16-224, with accuracy and *Macro-F1* scores exceeding or approaching 97%, markedly outperforming conventional CNNs and exhibiting strong long-tail learning capability that substantially enhances prediction accuracy for underrepresented rock classes; (2) On the lithological description generation task, PetroMind attains a BLEU-4 score of 0.644 and an SA-Rock score of 7.864, demonstrating exceptional few-shot learning performance

in multimodal contexts; (3) Performance on downstream geoscience tasks does not follow a simple positive correlation with model scale, which indicates that architectural design and task-aligned fine-tuning strategies are more decisive than parameter volume; and (4) Task-specific LoRA adapters prove more effective for high-resource tasks such as image classification, whereas a single shared LoRA adapter delivers superior multi-task interaction in low-resource captioning scenarios, likely due to cross-task knowledge transfer enabled by parameter sharing.

In the future, the instruction dataset will be expanded by annotating multi-scale rock images across a broader spectrum of lithologies, enhancing both data volume and geological diversity to improve captioning performance. In addition, in terms of task scope, knowledge extraction from rock description texts will be incorporated to extend the applicability of PetroMind.

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Data availability

Data will be made available on request.

Code availability section

Name of the code: PetroMind

Contact: Zhongliang Chen (c_mulder@163.com)

Hardware requirements: Codes and datasets were trained on a workstation equipped with two NVIDIA GeForce RTX 4090 GPUs and a linux container equipped with two NVIDIA A40 GPUs.

Program language: Python, Shell

Software required:

- **Windows workstation:** Python 3.11.11 with the PyTorch 2.6.0, PEFT 0.15.1, Torchvision 0.21.0, ModelScope 1.24.1 and sklearn 1.6.1

- **Linux container:** Python 3.11.5 with the PyTorch 2.1.1, PEFT 0.15.2, Torchvision 0.16.1, ModelScope

1.26.0 and sklearn 1.6.1

Program size: ~93.2 KB (without data)

The source codes are available for downloading at the link: <https://github.com/connamulder/PetroMind>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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